

Homework 1

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Problem 1. List all strategies in the extensive-form game.

Solution. A pure strategy specifies an action at every information set of a player, including information sets that are not reached in play.

Player 1 has one decision node, with actions A , B , and C . Thus,

$$S_1 = \{A, B, C\}.$$

Player 2 has two information sets: the node following A , and the information set joining the nodes following B and C . The dashed line means that player 2 cannot distinguish B from C , so the prescribed action must be the same at these two nodes.

Write a strategy of player 2 as (x_A, x_{BC}) , where the first component is the action after A and the second is the action at the shared information set after B or C . Then

$$S_2 = \{(a, a), (a, b), (b, a), (b, b)\}.$$

Hence player 1 has 3 pure strategies and player 2 has $2 \times 2 = 4$. If mixed strategies are included, their strategy spaces are $\Delta(S_1)$ and $\Delta(S_2)$, respectively, where $\Delta(S)$ denotes all probability distributions over S .

Problem 2. Best responses.

- (a) Against D , the payoffs from A, B, C are $1, 1, 0$.

Thus $\boxed{\text{BR}_1(D) = \{A, B\}}$. Every mixture of A and B is also optimal.

- (b) Let $r = \Pr(D) \in [0, 1]$. Player 1's expected payoffs are

$$u_1(A; r) = r, \quad u_1(B; r) = 2 - r, \quad u_1(C; r) = 2 - 2r.$$

Since $u_1(B; r) - u_1(A; r) = 2(1 - r)$ and $u_1(B; r) - u_1(C; r) = r$,

$$\boxed{\text{BR}_1(r) = \begin{cases} \{B, C\}, & r = 0, \\ \{B\}, & 0 < r < 1, \\ \{A, B\}, & r = 1. \end{cases}}$$

Mixed best responses are all mixtures supported on the listed sets. If

BR_1 means the pure strategies that are best responses to some belief,

then $\boxed{\text{BR}_1 = \{A, B, C\}}$.

- (c) For $p, q \geq 0, p + q \leq 1$, player 2's expected payoffs are

$$u_2(D; p, q) = 0p + q + 2(1 - p - q) = 2 - 2p - q,$$

$$u_2(E; p, q) = 2p + 0q + (1 - p - q) = 1 + p - q.$$

Since $u_2(D; p, q) - u_2(E; p, q) = 1 - 3p$,

$$\boxed{\text{BR}_2(p, q) = \begin{cases} \{D\}, & p < \frac{1}{3}, \\ \{D, E\}, & p = \frac{1}{3}, \\ \{E\}, & p > \frac{1}{3}. \end{cases}}$$

At $p = 1/3$, every mixture of D and E is optimal. The best response does not otherwise depend on q .

Problem 3. Strict dominance.

- (a) The strictly dominated pure strategy is $\boxed{B}$. Any strategy that puts no probability on B can be written as $\sigma_1 = \lambda A + (1 - \lambda)C$, with $\lambda \in [0, 1]$. Its payoffs against D and E are 3λ and $3(1 - \lambda)$, respectively. It strictly dominates B exactly when

$$3\lambda > 1 \quad \text{and} \quad 3(1 - \lambda) > 1, \quad \text{or equivalently} \quad \frac{1}{3} < \lambda < \frac{2}{3}.$$

Thus all the requested dominating strategies are

$$\boxed{\left\{ \lambda A + (1 - \lambda)C : \frac{1}{3} < \lambda < \frac{2}{3} \right\}}.$$

The endpoints yield equality against one column and hence do not give strict dominance. Neither A nor C is strictly dominated: A attains the maximum possible payoff 3 against D , and C attains the maximum possible payoff 3 against E .

- (b) Let τ_i strictly dominate σ_i . Allowing mixed strategies, define, for any $t \in (0, 1)$,

$$\rho_i^t = (1 - t)\sigma_i + t\tau_i.$$

Expected utility is linear in the player's own mixing probabilities, so for every s_{-i} ,

$$\begin{aligned} u_i(\rho_i^t, s_{-i}) - u_i(\sigma_i, s_{-i}) \\ = t[u_i(\tau_i, s_{-i}) - u_i(\sigma_i, s_{-i})] > 0. \end{aligned}$$

Therefore every ρ_i^t strictly dominates σ_i . Since $\tau_i \neq \sigma_i$, distinct $t \in (0, 1)$ give distinct strategies, proving that there are infinitely many such strategies. The inequalities also hold against mixed opponent strategies by taking expectations. $\square$

Problem 4. Beauty contests with actions in $[1, 2]$.

Solution. Let there be $n \geq 3$ players, as the question specifies many players. Write a for one's own action and $S = \sum_{j \neq i} a_j$ for the others' sum. Then $S \in [n-1, 2(n-1)]$ and $\bar{a} = (a + S)/n$.

- (a) With $u_i = -(a - \bar{a})^2$, **no pure strategy is strictly dominated**. To see this, fix any $a \in [1, 2]$ and suppose every other player chooses a . Choosing a then gives payoff 0, the largest possible payoff. Deviating to $b \neq a$ gives

$$-\left(b - \frac{b + (n-1)a}{n}\right)^2 = -\left(\frac{n-1}{n}\right)^2 (b-a)^2 < 0.$$

Consequently, no alternative pure or mixed strategy can give a strictly higher payoff against this opponents' action profile. Since this reasoning applies to every $a \in [1, 2]$, none is strictly dominated.

- (b) With $u_i = -(a - 2\bar{a})^2$, **all pure strategies in $[1, 2)$ are strictly dominated by 2**. Holding the other players' actions fixed, the payoff is

$$u_i(a, S) = -\frac{((n-2)a - 2S)^2}{n^2}.$$

For $n > 2$, this payoff is strictly increasing in a on $[1, 2]$, because

$$\frac{\partial u_i(a, S)}{\partial a} = \frac{2(n-2)}{n^2} [2S - (n-2)a] > 0,$$

where $2S - (n-2)a \geq 2(n-1) - 2(n-2) = 2 > 0$. Thus $u_i(2, S) > u_i(a, S)$ for every $a < 2$ and every feasible S . Action 2 is the unique best response to every opponents' profile, so it cannot be strictly dominated. The requested set is therefore

$$\boxed{[1, 2].}$$